

RISHABH KUMAR

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TECHNICAL SKILLS

Languages: C, C++, Rust, JavaScript, TypeScript

Web & Backend: Next.js, React, Node.js, Express, MongoDB (MERN Stack)

Systems & Infra: Linux, Docker, Redis, Git, OpenGL, Raylib

Concepts: REST APIs, Containerization

EDUCATION

Indian Institute of Technology Madras

2025 – Present

BS Degree Program (Online) — Data Science & Applications GPA: **7.5/10**

BIT Sindri

2025 – Present

B.Tech in Civil Engineering

PROFESSIONAL EXPERIENCE

Freelance Software Developer

2025 – Present

- Designed and developed a production company website for a client, handling end-to-end delivery including UI design, frontend implementation, deployment, and post-launch maintenance.
- Delivered a performant, responsive web presence using modern JavaScript frameworks, coordinating directly with stakeholders to meet design and functional requirements.

KEY PROJECTS

Sketch.ai — Visual Learning Platform | *Next.js, LLM APIs, Excalidraw* 🐙 sketch.ai

- Engineered a visual learning platform integrating Excalidraw with large language models to dynamically generate diagrams, animations, and flowcharts for physics, mathematics, and engineering concepts.
- Designed a prompt pipeline that translates natural language queries into structured visual representations with step-by-step animation support.
- Architected a modular frontend enabling real-time collaborative whiteboard sessions with AI-generated overlays.

Minecraft Engine in C — Voxel Renderer | *C, OpenGL, GLFW*

- Built a fully functional Minecraft-inspired voxel engine from scratch in C using OpenGL, implementing chunk-based world generation, frustum culling, and a first-person camera system.
- Engineered low-level graphics pipelines including VAO/VBO management, shader compilation, texture atlasing, and real-time face culling to optimize render performance.
- Implemented a voxel world data structure with efficient memory layout to support large dynamic worlds at interactive frame rates.

Break the Code — LLM Debugging Arena | *Next.js, Docker, LLM APIs*

- Developed an interactive platform where users intentionally introduce bugs into code while an LLM attempts to identify and repair them in real time.
- Containerized sandboxed code execution environments using Docker to ensure secure, isolated, and scalable runtime evaluation of user-submitted code.
- Built a full-stack Next.js application with API routes managing LLM inference calls, session state, and sandboxed process lifecycle management.

Sorting Algorithm Visualizer | *C, Raylib*

- Implemented a real-time algorithm visualization engine in C using Raylib, rendering step-by-step execution of sorting algorithms including merge sort, quicksort, and heapsort.
- Designed a frame-rate-controlled animation loop decoupling visual rendering from algorithmic state transitions to accurately reflect computational complexity behavior.

COMPETITIVE PROGRAMMING

Codeforces: Max Rating **950+** | Solved **100+** algorithmic problems across graphs, greedy algorithms, and data structures

LeetCode: Solved **100+** coding problems focused on DSA and problem-solving patterns